Arithmetic-Geometric Structure of Agrawal's Conjecture and the Torus Dimension Count Behind the Double Condition

(AG track working document — joint correctness audit)

Dan Shumow Claude (Anthropic Fable 5)

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Joint audit notice (22 October 2026). This source corrects the fixed-ring dictionary, support hypothesis and global versus finite-field distinctions. The elliptic section now proves only its cofactor upper bound; the function-field appendix is exploratory. Historical N1/A10 experiments do not establish an algorithmic no-go theorem. The same-name PDF beside this source is historical; the rebuilt audit version is in `doc/audit-pdfs/`. Current status and unresolved modeling questions are in `doc/integration-review-status.md`.

Abstract

We study the cyclotomic support bounds of Frobenius-matching constructions related to Agrawal's conjecture. For even $d = \mathrm{ord}_r(p)$ the exact identity $(X-1)^{p^{d/2}-1} = -X^{-1}$ gives $\rho_r(p) \mid 2r(p^{d/2}-1)$. The associated containing torus has dimension $d' = d/2$ for even d and $d' = d$ for odd d. Valid Frobenius matching forces $d \geq 3$, so this model's minimum dimension is 2, at $d = 4$. Turning the upper support bound into a necessary smoothness condition requires an explicit, section-specific support-sharpness hypothesis; neither its density nor the cost of targeting exceptional primes is established here.

For Popovych's section $X+2$, a characteristic-zero ideal argument rules out Galois-conjugate ratios that are roots of unity. It does not exclude accidental finite-field collapses or prove generic order saturation. Under a separate full-support hypothesis, the pair has model dimension $\min\{d : d \mid r-1,\ d \geq 3\}$, equal to 4 at $r = 5$. Dickman estimates are parameter-specific heuristics, not harvest impossibility theorems. We also record the generalized-Carmichael dictionary, the automatic $X+1$ congruence under matching plus the Carmichael condition, and a cofactor-based stabilizer-index upper bound. The stronger elliptic lower bound, the transfer of solver obstructions, and unconditional function-field sharpness/existence claims are not established. The original numerical artifacts are preserved as finite checks; they do not certify these broader conclusions or settle either conjecture.

1 The Frobenius-lift dictionary

This section fixes the geometric reading used throughout (item AG1). Write $T(-1, n, r)$ for the AKS congruence

$$(X-1)^n \equiv X^n - 1 \pmod{n,\ X^r - 1}, \tag{1}$$

r an odd prime not dividing n.

Proposition 1.1 (primality as a Frobenius lift)**.** *Let $A_n = (\mathbb{Z}/n)[X]/(X^r-1)$, the mod-$n$ group algebra of μ_r, and let $\sigma_n : X \mapsto X^n$ be the n-th Adams operation (well defined since $X^r = 1$). If n is prime, then the n-power map on A_n coincides with σ_n; that is, $f(X)^n \equiv f(X^n) \pmod{n,\ X^r-1}$ for* every *f. The congruence (1) is this coincidence tested on the single element $f = X-1$.*

Proof. For n prime the n-power map is a ring endomorphism (freshman's dream) and fixes $\mathbb{Z}/n$ (Fermat), so $f(X)^n \equiv f^{(n)}(X^n) \equiv f(X^n)$, where $f^{(n)}$ raises coefficients to the n-th power. □

Before truncation, the coincidence characterizes primality:

Proposition 1.2 (the untruncated test [1, Lemma 2.1])**.** *Let a be coprime to $n \geq 2$. Then n is prime if and only if $(X+a)^n \equiv X^n + a \pmod{n}$ in $(\mathbb{Z}/n)[X]$. (If $q^k \parallel n$ for a prime $q < n$, the coefficient of X^q is $\binom{n}{q} a^{n-q} \not\equiv 0 \bmod n$.) The AKS test, and Agrawal's conjecture, concern how much of this characterization survives the truncation from the affine line to the finite subscheme μ_r.*

In λ-ring language the statement is canonical rather than ad hoc. $\mathbb{Z}[X]/(X^r - 1) = \mathbb{Z}[\mathbb{Z}/r]$ carries the Adams operations $\psi^k : X \mapsto X^k$, which pairwise commute, and each ψ^p is a *Frobenius lift*: $\psi^p(f) \equiv f^p \pmod{p}$ for every f (the calculation of Proposition 1.1). By Wilkerson's criterion [12], a torsion-free ring with such a commuting family of Frobenius lifts carries a unique λ-ring structure; in Borger's absolute geometry a Λ-structure *is* descent data to $\mathbb{F}_1$ [11]. In this language:

$$n \text{ is prime} \Longrightarrow \psi^n(f) \equiv f^n \pmod{n} \quad \text{for every } f.$$

The converse fails in a fixed truncated ring: take $n = 399001 = 31 \cdot 61 \cdot 211$ and $r = 5$. Each prime factor is 1 mod 5 and each $p-1$ divides $n-1$. Thus A_n is a product of copies of these prime fields, every element satisfies $f^n = f$, and ψ^n is the identity since $n \equiv 1 \pmod{5}$. This composite passes the coincidence on *every* section, inside the conjecture's exception clause. Wilkerson's criterion does not assert the converse. The AKS test is the coincidence on one section; its proposed primality consequence requires the exception clause and the other algorithmic safeguards. The pseudofield constructions of Lenstra–Pomerance [3] provide related Frobenius data, not an identification of all these truncated tests with certified pseudofields.

Lemma 1.3 (the exceptional clause)**.** *$n^2 \equiv 1 \pmod{r}$ iff σ_n, viewed as the element $n \bmod r$ of $(\mathbb{Z}/r)^\times \cong \mathrm{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q})$, lies in $\mathrm{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q}(\zeta_r)^+) = \{1, \text{complex conjugation}\}$.*

Proof. $(\mathbb{Z}/r)^\times$ is cyclic of even order $r-1$; its unique element of order 2 is -1, which acts on μ_r as inversion, i.e. as complex conjugation, the generator of $\mathrm{Gal}(\mathbb{Q}(\zeta_r)/\mathbb{Q}(\zeta_r)^+)$. □

Agrawal's conjecture thus asserts: *a composite Frobenius impostor on the single section $X-1$ must act trivially on the maximal real subfield.* A *valid counterexample* is a composite n with $T(-1, n, r)$ and $t := \mathrm{ord}_r(n) > 2$.

2 Setup: cyclotomic tori and the mechanism

2.1 Cyclotomic tori

For $e \geq 1$ let Φ_e denote the e-th cyclotomic polynomial, so that

$$p^d - 1 \;=\; \prod_{e \mid d} \Phi_e(p). \tag{2}$$

Over $\mathbb{F}_p$, the Weil restriction $\mathrm{Res}_{\mathbb{F}_{p^d}/\mathbb{F}_p} \mathbb{G}_m$ decomposes up to isogeny as $\prod_{e|d} T_e$, where T_e is the cyclotomic torus of dimension $\varphi(e)$ split by $\mathbb{F}_{p^e}$, with $\#T_e(\mathbb{F}_p) = \Phi_e(p)$ [8, 9]. The two *one-dimensional* tori are $T_1 = \mathbb{G}_m$ ($\#T_1(\mathbb{F}_p) = p-1$, the split torus) and T_2 the norm-one torus of $\mathbb{F}_{p^2}/\mathbb{F}_p$ ($\#T_2(\mathbb{F}_p) = p+1$, the nonsplit twist); $\varphi(e) = 1$ iff $e \in \{1, 2\}$. We refer to the index e in (2) as the *level*.

2.2 The order of $X-1$

Fix primes $p \neq r$ and let $d = d_p := \mathrm{ord}_r(p)$. Then

$$\mathbb{F}_p[X]/(X^r-1) \;\cong\; \mathbb{F}_p \times \mathbb{F}_p[X]/(\Phi_r(X)), \qquad \mathbb{F}_p[X]/(\Phi_r(X)) \;\cong\; \prod_{i=1}^{(r-1)/d} \mathbb{F}_{p^d},$$

the factors indexed by the orbits of $\langle p \rangle$ on $(\mathbb{Z}/r)^\times$; in the i-th factor X maps to a primitive r-th root of unity ζ_i. The element $X-1$ vanishes in the $\mathbb{F}_p$ factor (where (1) holds trivially: both sides are 0) and is a unit in $\mathbb{F}_p[X]/\Phi_r$ since $\Phi_r(1) = r \neq 0$ in $\mathbb{F}_p$.

Definition 2.1. *$\rho_r(p)$ is the multiplicative order of the image of $X-1$ in $\big(\mathbb{F}_p[X]/\Phi_r(X)\big)^\times$, i.e. $\rho_r(p) = \mathrm{lcm}_i \,\mathrm{ord}(\zeta_i - 1)$.*

For $r = 5$ this is the quantity $\rho(p)$ of Váňa [7], with $\rho(p) \mid 10(p^2-1)$; Corollary 3.3 below recovers this.

2.3 The Lenstra–Pomerance mechanism

Definition 2.2 (L–P mechanism)**.** *A squarefree composite $n = p_1 \cdots p_k$ coprime to r is produced by the* L–P mechanism *if for every i there is $j_i \geq 0$ with*

$$n \;\equiv\; p_i^{j_i} \pmod{\mathrm{lcm}\big(r,\ \rho_r(p_i)\big)}, \tag{3}$$

and $\mathrm{ord}_r(n) > 2$ (validity).

The original $r = 5$ construction [2, 4] uses a uniform exponent $j_i = j \in \{1, 3\}$ with $j \equiv k \pmod 4$: $j = 1$ for $k \equiv 1 \pmod 4$ and $j = 3$ for $k \equiv 3 \pmod 4$. The conditions $p \equiv 3 \pmod{80}$, $p-1 \mid n-1$ and $p+1 \mid n+1$ give $n \equiv p^j \pmod{10(p^2-1)}$, by CRT on the pairwise-coprime factors 80, $(p-1)/2$ and $(p+1)/4$. With $\rho_5(p) \mid 10(p^2-1)$ this implies $n \equiv p^j \pmod{\mathrm{lcm}(5, \rho_5(p))}$ in both cases. Our Track A pipeline (`PLAN.md`) is a search inside Definition 2.2.

Remark 2.3 (scope). The mechanism is sufficient, not necessary, for (1): the component equation $(\zeta-1)^n = \zeta^n - 1$ need not be analyzed here outside the Frobenius-matching family (3). The original L–P sufficient construction is in this family. We do not prove necessity, classify all possible constructions, or exclude accidental local solutions.

3 The torus dimension count

3.1 Frobenius matching and the inversion collapse

Lemma 3.1 (Frobenius matching)**.** *With notation as above:*

(i) $(X-1)^{p^k} \equiv X^{p^k} - 1 \pmod{p,\ X^r - 1}$ *for all $k \geq 0$.*

(ii) If $n \equiv p^j \pmod{\mathrm{lcm}(r, \rho_r(p))}$ *then $T(-1, n, r)$ holds modulo p.*

(iii) Conversely, (3) *forces* $n \equiv p^j \pmod r$*, hence* $\mathrm{ord}_r(n) \mid d_p$*. In particular a valid counterexample (*$\mathrm{ord}_r(n) > 2$*) forces $d_p \geq 3$ for every $p \mid n$.*

Proof. (i) is the iterated Frobenius endomorphism of $\mathbb{F}_p$-algebras. (ii): write $n = p^j + c\cdot\mathrm{lcm}(r, \rho_r(p))$. In the $\mathbb{F}_p$ factor both sides of (1) vanish. In $\mathbb{F}_p[X]/\Phi_r$: $(X-1)^n = (X-1)^{p^j}\left((X-1)^{\rho_r(p)}\right)^{c'} = (X-1)^{p^j} \equiv X^{p^j} - 1$ by (i), and $X^{p^j} = X^{n \bmod r} = X^n$ since $X^r \equiv 1 \bmod \Phi_r$ (as $X^r - 1 = (X-1)\Phi_r(X)$ for r prime). (iii): reduce (3) modulo r; then $n \in \langle p\rangle \subseteq (\mathbb{Z}/r)^\times$, so $\mathrm{ord}_r(n)$ divides $\#\langle p\rangle = d_p$, and $\mathrm{ord}_r(n) \geq 3$ gives $d_p \geq 3$. □

Lemma 3.2 (inversion collapse)**.** *Suppose $d = \mathrm{ord}_r(p)$ is even. Then, in $\mathbb{F}_p[X]/\Phi_r(X)$,*

$$(X-1)^{p^{d/2}-1} \;=\; -X^{-1},$$

and consequently

$$\rho_r(p) \;\Big|\; 2r\,(p^{d/2}-1) \;=\; 2r \prod_{e \mid d/2} \Phi_e(p). \tag{4}$$

Proof. $(\mathbb{Z}/r)^\times$ is cyclic, so its unique element of order 2 is -1; since $\mathrm{ord}_r(p) = d$, the element $p^{d/2}$ has order 2, whence $p^{d/2} \equiv -1 \pmod r$. In each field component, with $X \mapsto \zeta$ a primitive r-th root of unity, iterated Frobenius gives

$$(\zeta-1)^{p^{d/2}} = \zeta^{p^{d/2}} - 1 = \zeta^{-1} - 1 = -\zeta^{-1}(\zeta - 1),$$

and dividing by $\zeta - 1 \neq 0$ yields $(\zeta-1)^{p^{d/2}-1} = -\zeta^{-1}$. This holds in every component, hence in the product ring. Finally $(-X^{-1})^{2r} = X^{-2r} = 1$, so $(X-1)^{2r(p^{d/2}-1)} = 1$, giving (4); the product form is (2) applied to $p^{d/2} - 1$. □

Corollary 3.3 (Váňa's bound)**.** *For $r = 5$ and $p \equiv \pm 2 \pmod 5$ (so $d = 4$), $\rho_5(p) \mid 10(p^2-1)$.*

Definition 3.4 (support level)**.** *Set*

$$d' \;=\; d'(p) \;:=\; \begin{cases} d/2, & d \text{ even}, \\ d, & d \text{ odd}. \end{cases}$$

For odd d the trivial bound $\rho_r(p) \mid p^d - 1 = \prod_{e\mid d}\Phi_e(p)$ applies (orders in a product of copies of $\mathbb{F}_{p^d}^\times$), so in both cases

$$\rho_r(p) \;\Big|\; 2r \prod_{e \mid d'} \Phi_e(p)\,: \tag{5}$$

the p-dependent part of $\rho_r(p)$ is supported on the cyclotomic levels $e \mid d'$.

3.2 Generic sharpness

The bound (5) is an upper bound on support; the mechanism's cost analysis also needs it to be generically *attained* — otherwise a harvest could fish for primes p where $\rho_r(p)$ collapses further.

Hypothesis 3.5 (support sharpness at a specified threshold)**.** *Put $P_r(p) = \prod_{e\mid d'}\Phi_e(p)$ and fix $B \geq 2rd$. For the selected prime p, assume every prime $\ell > B$ dividing $P_r(p)$ also divides $\rho_r(p)$. This is a hypothesis about prime support, not divisibility with full prime-power multiplicities. For another section with order $\rho_f(p)$ and a permitted product $P_f(p)$, the analogous assumption must be stated separately.*

Remark 3.6 (what the uniform model actually says). In a cyclic group of order N with $a = v_\ell(N)$, one has $\ell \nmid \mathrm{ord}(x)$ iff x is an ℓ^a-th power: writing $x = g^b$, both say $\ell^a \mid b$. A uniform element misses ℓ with probability ℓ^{-a}, not necessarily $1/\ell$. Being merely an ℓ-th power does not suffice when $a > 1$. Independent uniform components would multiply these probabilities, but the values $\zeta_i - 1$ have algebraic relations and are not proved to follow that distribution. The committed AG2 samples are finite sanity checks, not a density theorem or a quantified comparison with the twin-smooth population.

3.3 The dimension-count theorem

Theorem 3.7 (torus dimension count)**.** *Let $r \geq 5$ be prime and let an odd n be produced by the L–P mechanism (Definition 2.2) with $\mathrm{ord}_r(n) > 2$. For each $p \mid n$ put $d_p = \mathrm{ord}_r(p)$ and d'_p as in Definition 3.4. Assume Hypothesis 3.5 at every $p \mid n$, with one common $B \geq \max_{p\mid n} 2rd_p$. In the design requiring a B-smooth matching modulus, $\rho_r(p)$ is B-smooth if and only if:*

$$\Phi_e(p) \text{ is } B\text{-smooth for every } e \mid d'_p \quad \Longleftrightarrow \quad \#\Big(\textstyle\prod_{e\mid d'_p} T_e\Big)(\mathbb{F}_p) \text{ is } B\text{-smooth},$$

a smoothness condition on the points of a containing *torus of dimension*

$$\dim \prod_{e\mid d'_p} T_e \;=\; \sum_{e\mid d'_p} \varphi(e) \;=\; d'_p \;=\; \begin{cases} d_p/2, & d_p \text{ even}, \\ d_p, & d_p \text{ odd}. \end{cases}$$

Moreover:

(a) $d'_p \geq 2$ *for every $p \mid n$: the burden is never a single one-dimensional torus. (Indeed $d'_p = 1$ would force $d_p \in \{1, 2\}$, contradicting $d_p \geq 3$ from Lemma 3.1(iii).)*

(b) $d'_p = 2$ *iff $d_p = 4$, in which case the level set is $\{1, 2\}$ and the condition is: $p - 1 = \#T_1(\mathbb{F}_p)$ and $p + 1 = \#T_2(\mathbb{F}_p)$ both B-smooth, i.e. twin-smoothness of $\big(\frac{p-1}{2}, \frac{p+1}{2}\big)$. This is the supply condition used in the original Carmichael/Lucas–Carmichael construction, not by itself a proof of either divisibility condition on n.*

(c) $d_p = 4$ *is realizable iff $4 \mid r - 1$; hence for $r \equiv 1 \pmod 4$ the minimal burden is 2, identical in structure to the $r = 5$ case, while for $r \equiv 3 \pmod 4$ every admissible d_p gives burden ≥ 3.*

(d) The minimal burden as a function of r is

$$\mathrm{burden}(r) \;=\; \min_{\substack{d\mid r-1 \\ d\geq 3}} \begin{cases} d/2, & d \text{ even} \\ d, & d \text{ odd} \end{cases}$$

which reproduces the measured A11 table for all eleven moduli tested there (Section 4).

Proof. If $P_r(p)$ is B-smooth then so is $\rho_r(p)$ by (5) and $2r \leq B$. Conversely Hypothesis 3.5 implies every prime above B dividing $P_r(p)$ would divide $\rho_r(p)$. This proves the smoothness equivalence. The dimension computation is $\sum_{e\mid m} \varphi(e) = m$ with $m = d'_p$, together with $\dim T_e = \varphi(e)$ and $\#T_e(\mathbb{F}_p) = \Phi_e(p)$ (Section 2). (a): $d_p \geq 3$ by Lemma 3.1(iii); $d'_p = 1$ requires $d_p \leq 2$. (b): $d'_p = 2$ with d_p even means $d_p = 4$ and levels $\{e : e \mid 2\} = \{1, 2\}$; $d'_p = 2$ with d_p odd is impossible ($d_p = 2$ is even). With p odd, B-smoothness of $p \mp 1 = 2m, 2(m+1)$ is twin-smoothness of the pair $(m, m+1)$ up to the factor 2. (c): $d_p = 4$ requires $4 \mid \#(\mathbb{Z}/r)^\times = r - 1$; conversely if $4 \mid r - 1$ the cyclic group

$(\mathbb{Z}/r)^\times$ has elements of order 4, and primes p in those residue classes exist (Dirichlet). This realizes the local order $d_p = 4$, not the assembly of a composite counterexample. For $r \equiv 3 \pmod 4$: even $d \mid r-1$ satisfy $d \equiv 2 \pmod 4$, so even $d \geq 3$ means $d \geq 6$, burden $d/2 \geq 3$; odd $d \geq 3$ gives burden $d \geq 3$. (d) is the minimization of d'_p over admissible d_p. □

Corollary 3.8 (conditional minimum in the support model)**.** *Under the hypotheses and smooth-modulus design of Theorem 3.7, the containing-torus dimension is at least* 2*. Its unique dimension-*2 *level set is* $\{1, 2\}$*, locally realizable for* $r \equiv 1 \pmod 4$*. This is not an unconditional lower bound on the actual order at every prime, a runtime lower bound, or a closure of all alternative-modulus constructions.*

Remark 3.9 (cost interpretation). Under the standard Dickman heuristic, B-smoothness of $\Phi_e(p)$, a quantity of size $p^{\varphi(e)}$, has cost governed by $u = \varphi(e) \log p / \log B$; total burden across the level set is modeled using $\sum_{e \mid d'} \varphi(e) = d'$. Polynomial-value smoothness, local congruence biases and joint dependence still matter: total dimension alone does not determine harvest yield or solver cost. The project has sampled twin-smooth supply, not proved that every higher-dimensional construction is more expensive. Torus-based cryptography [9] provides related group structures, not a computational lower bound for this search.

Remark 3.10 (what remains conditional). Lemmas 3.1 and 3.2, the upper bound (5), and the dimension minimization for the chosen containing tori are unconditional. Necessity of the smoothness conditions uses Hypothesis 3.5. Neither an asymptotic density for its failures nor a comparison with smooth-prime supply has been proved here. Targeting exceptional orders is not excluded by this argument.

4 Numerical validation

`code/ag2_torus_check.py` (deterministic; output in `data/generated/ag2_torus_check.json`) verifies 67 sample primes spanning

$$(r, d) \in \{(5,4), (13,4), (41,4), (7,3), (7,6), (19,3), (11,5), (23,11)\}.$$

These were sampled independently of the A11 scan, which drew $p > 1000$ in one fixed residue class per r, whereas here p runs from 3 upward over all classes of the given order, and the implementation is independent (exact orders by polynomial arithmetic in $\mathbb{F}_p[X]/\Phi_r$ rather than A11's level attribution):

check	statement	result
V1	Frobenius matching (Lemma 3.1(i))	67/67
V2	collapse identity $(X-1)^{p^{d/2}-1} = -X^{-1}$ (Lemma 3.2)	36/36
V3	exact $\rho_r(p)$ computed within bound (5)	67/67
V4	sampled large primes of levels $e \mid d'$ divide $\rho_r(p)$ (finite check, not Hypothesis 3.5)	37/37
V5	top level $e = d'$ genuinely needed in $\rho_r(p)$	67/67
V6	burden formula (Thm. 3.7(d)) vs. A11 table	11/11

These are historical finite checks; they do not prove support saturation. V4's 0 exceptions were compared to the historical $1/\ell$ model (≈ 0.19 misses expected over this sample). V6 reproduces both the burden value and minimizing d of the A11 measurement for

$$r \in \{5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41\}$$

(`data/FINDINGS.md`, 2026-08-24).

5 A quantified heuristic for Popovych's conjecture

Popovych's conjecture [4] strengthens the hypothesis of Agrawal's to the simultaneous pair

$$(X-1)^n \equiv X^n - 1 \quad \text{and} \quad (X+2)^n \equiv X^n + 2 \pmod{n,\, X^r - 1}, \tag{6}$$

with the same conclusion. This section compares support models for the Frobenius-matching mechanism. The first section has the exact inversion identity of Lemma 3.2; the second lacks its characteristic-zero analogue. A larger necessary smoothness burden still requires the explicit section-specific support hypotheses. No priority claim or proof of resistance to all counterexample constructions is made here.

Throughout, $r \neq 3$ is prime and $p \nmid 3(2^r+1)$; the finitely many excluded p are degenerate ($p = 3$ makes $-2 \equiv 1$, so the two sections of (6) *coincide* mod 3; $p \mid (2^r+1)/3$ makes $X+2$ a zero divisor in some component).

Lemma 5.1 (the level-one component is base-3 Fermat)**.** *The $X = 1$ component of the second congruence in (6) is $3^n \equiv 3 \pmod{n}$. In particular it is automatic for any Carmichael-type n ($p - 1 \mid n - 1$ for all $p \mid n$), hence free for the L–P construction; but unlike the first section, whose $X = 1$ component is $0 = 0$, the second section is already nontrivial at level one.*

Proof. Evaluate (6) at $X = 1$: $(1+2)^n \equiv 1 + 2 \pmod{n}$. □

Lemma 5.2 (global independence)**.** *In $\mathbb{Z}[\zeta_r]$ the elements $\zeta - 1$ and $\zeta + 2$ have principal ideals with disjoint prime support: $N(\zeta - 1) = \Phi_r(1) = r$ while $N(\zeta + 2) = \Phi_r(-2) = (2^r+1)/3$, and $\gcd\big(r,\, 2^r + 1\big) = \gcd(r, 3) = 1$. Since neither is a unit, $\zeta - 1$ and $\zeta + 2$ generate a rank-2 subgroup of $\mathbb{Q}(\zeta_r)^\times$ modulo roots of unity. This is global multiplicative independence; it does not assert independence of finite-field orders.*

Proof. $X^r - 1 = (X-1)\Phi_r(X)$ gives $\Phi_r(-2) = ((-2)^r - 1)/(-3) = (2^r+1)/3$. By Fermat $2^r \equiv 2 \pmod{r}$, so $2^r + 1 \equiv 3 \pmod{r}$ and $\gcd(r, 2^r+1) = \gcd(r, 3) = 1$ as $r \neq 3$. Disjoint support plus non-unit norms gives independence modulo torsion. □

Lemma 5.3 (global non-unit ratios)**.** *For every $\gamma \in (\mathbb{Z}/r)^\times$ with $\gamma \neq 1$, the ratio $\sigma_\gamma(\zeta+2)/(\zeta+2) = (\zeta^\gamma+2)/(\zeta+2)$ is not a root of unity — it is not even a unit of $\mathbb{Z}[\zeta_r]$. Thus there is no characteristic-zero root-of-unity identity for this ratio analogous to the inversion identity for $X - 1$. This does not exclude special reductions.*

Proof. Let $\mathfrak{q}$ be a prime of $\mathbb{Z}[\zeta_r]$ dividing $(\zeta + 2)$, so $\zeta \equiv -2 \pmod{\mathfrak{q}}$ and the image of -2 in the residue field has multiplicative order r (it is the image of ζ). If $\mathfrak{q}$ also divided $(\zeta^\gamma + 2)$ then $\zeta^\gamma \equiv -2 \equiv \zeta$, so $\zeta^{\gamma-1} \equiv 1 \pmod{\mathfrak{q}}$, forcing $r \mid \gamma - 1$, i.e. $\gamma = 1$. So for $\gamma \neq 1$ the ratio has nonzero valuation at $\mathfrak{q}$ and is not a unit.

Contrast with the first section: $\sigma_{-1}(\zeta - 1)/(\zeta - 1) = -\zeta^{-1} \in \mu_{2r}$, available because the ideal $(\zeta - 1)$ (the totally ramified prime above r) is fixed by the whole Galois group — the ratio under inversion is torsion for $X - 1$ but not for $X + 2$. This statement concerns conjugate ratios at ζ, not the absolute Galois stability of the rational point -2. □

Remark 5.4 (finite-field exception). For $r = 13$, $p = 5$, one has $5^2 \not\equiv 1 \pmod{13}$ but $(X+2)^{24} \in \mu_{26}$ in $\mathbb{F}_5[X]/\Phi_{13}(X)$. This is not excluded by $p \nmid 3r(2^r+1)$. The global non-unit proof does not prevent such accidental finite-field relations.

Corollary 5.5 (conditional smoothness for the second section)**.** *Define $\rho'_r(p)$ as the multiplicative order of $X+2$ in $(\mathbb{F}_p[X]/\Phi_r(X))^\times$. Then $\rho'_r(p) \mid p^d-1 = \prod_{e\mid d}\Phi_e(p)$. If the section-specific version of Hypothesis 3.5 holds with $P_f(p) = p^d-1$ and $B \geq 2rd$, then $\rho'_r(p)$ is B-smooth iff every $\Phi_e(p)$, $e \mid d$, is B-smooth. The global lemma does not prove that support hypothesis.*

Theorem 5.6 (Popovych burden)**.** *Run the L–P mechanism for the pair (6): for each $p \mid n$,*

$$n \equiv p^{j_p} \pmod{\operatorname{lcm}\big(r,\ \rho_r(p),\ \rho'_r(p)\big)}, \qquad \operatorname{ord}_r(n) > 2.$$

Assume the section-specific support hypotheses above for both sections at a common $B \geq \max_{p\mid n} 2rd_p$. In the B-smooth matching-modulus design, the pair then requires every $\Phi_e(p)$, $e \mid d_p$, to be B-smooth. The dimension of this containing torus is $\sum_{e\mid d_p}\varphi(e) = d_p$ for every *parity of d_p. Hence*

$$\operatorname{burden}_{\mathrm{Pop}}(r) \;=\; \min\{\, d : d \mid r-1,\ d \geq 3 \,\} \;\geq\; 3 \qquad \textit{for every prime } r,$$

and:

(a) the global optimum is 3 (attained iff $3 \mid r-1$), versus 2 for the single section (Theorem 3.7);

(b) at $r = 5$ — Popovych's own proposed modulus — and generally whenever $4 \mid r-1$ and $3 \nmid r-1$, the burden is 4: the pool primes must have $p-1$, $p+1$ and p^2+1 simultaneously smooth;

(c) $\operatorname{burden}_{\mathrm{Pop}}(r) \geq \operatorname{burden}_{\mathrm{Agr}}(r)$ for every r, strictly whenever $\operatorname{burden}_{\mathrm{Agr}}(r) = 2$ (i.e. for every $r \equiv 1 \pmod 4$);

(d) the twin-smooth regime (burden 2) — the only regime this project found supply-feasible (gates G1/G2) — does not exist for the pair at any modulus.

Proof. Corollary 5.5 gives the smoothness equivalence for the second section under its own support hypothesis, not from global independence alone. Its containing product has dimension $\sum_{e\mid d}\varphi(e) = d$. The single-section product has dimension d' by Theorem 3.7. Admissibility $d \geq 3$ is Lemma 3.1(iii), unchanged. Minimizing d over $d \mid r-1$, $d \geq 3$ gives the formula. (a): the least divisor ≥ 3 of $r-1$ is 3 iff $3 \mid r-1$. (b): if $3 \nmid r-1$ and $4 \mid r-1$ the least is 4, with levels $\{1,2,4\}$ and $\Phi_4(p) = p^2+1$. (c): $\operatorname{burden}_{\mathrm{Agr}}(r)$ minimizes d' where $d' \leq d$ always, so it is $\leq \operatorname{burden}_{\mathrm{Pop}}(r)$; when it equals 2 the Popovych value is ≥ 3 by the displayed bound. (d) is (a) plus Theorem 3.7(b). □

Remark 5.7 (heuristic cost at the projected scale). At the frozen harvest spec of this project (`doc/harvest-spec.md`: pool primes $p \approx 51$ bits, projected supply/demand ratio 13.8, pool 2.5×10^5), the extra quadratic level of Theorem 5.6(b) is a pure Dickman tax on a ≈ 102-bit quantity: $\rho(u)$ with $u = 102/\log_2 B$, i.e.

$$u \approx 9.9,\ \rho(u) \approx 3.9 \times 10^{-11} \quad (B = 1259), \qquad u \approx 7.5,\ \rho(u) \approx 1.8 \times 10^{-7} \quad (B' = 12590),$$

multiplying the projected ratio down to 5×10^{-10}–2.5×10^{-6} and the expected pool to 10^{-5}–0.05 elements under this model. This additionally assumes Dickman behavior for the polynomial values and the relevant conditional independence. It is not a statement that every feasible harvest is empty, nor does it exclude structured enumeration or exceptional-order designs. The N1/A10 experiments do not establish a solver obstruction that transfers to the pair, and their density parameters are not measurements of full H7 component identities.

Remark 5.8 (interpretation and scope). Theorem 5.6 compares the containing dimensions under its support hypotheses: global minima 3 versus 2, and 4 versus 2 at $r = 5$. Remark 5.7 adds a separate heuristic cost model. Neither is a proof about every pool or construction, a universal solver lower bound, or evidence deciding the truth of either conjecture. Modulo $p = 3$ the sections coincide, an additional degeneracy that must be handled explicitly.

Remark 5.9 (the constant 2 is immaterial). For any integer $c \notin \{0, \pm 1\}$ the argument of Lemma 5.3 runs verbatim for the section $X + c$: at a prime $\mathfrak{q} \mid (\zeta + c)$ the image of $-c$ has multiplicative order r, so no $\gamma \neq 1$ preserves the divisor, the stabilizer is trivial. With a separate support-sharpness hypothesis for that section, the conditional burden calculation of Theorem 5.6 applies, with c entering only through the norm $\Phi_r(-c) = (c^r + 1)/(c + 1)$ (whose primes, with $p \mid c(c + 1)$, are the finitely many excluded p), the excluded modulus $r \mid c + 1$ (for $c = 2$: $r = 3$), and the level-one component, the base-$(c+1)$ Fermat condition. What must be avoided is exactly degeneracy of the zero $-c$: it must be a point of $\mathbb{G}_m$ avoiding the rational torsion $\mu_2(\mathbb{Q}) = \{\pm 1\}$, i.e. $c \notin \{0, \pm 1\}$. The exclusion $c = 0$ is vacuous rather than subtle (X is a monomial, so $(X)^n \equiv X^n$ is an identity in every ring; geometrically the zero sits at the origin, off the torus, and the section is a unit of finite order r). Among nonzero c: $c = -1$ is the first section itself, and $c = +1$ — the one innocent-looking bad choice — has its zero at -1, fixed by inversion, so the collapse *survives*: $(\zeta + 1)^{p^{d/2}} = \zeta^{-1}(\zeta + 1)$, giving $\mathrm{ord}(X + 1) \mid r(p^{d/2} - 1)$ for even d; worse, $\zeta + 1 = (\zeta^2 - 1)/(\zeta - 1)$ is multiplicatively dependent on the Galois orbit of $\zeta - 1$, so the pair $\{X - 1, X + 1\}$ adds no rank and no burden at all — a Popovych-style conjecture with $c = 1$ would be no harder for the mechanism than Agrawal's. In the language of Theorem 7.3: $U(X + c) = \{\pm 1\}$ for $c = \pm 1$ and $\{1\}$ otherwise; $c = 2$ is simply the smallest valid choice ($c = -2$ works equally, with Mersenne norm $2^r - 1$). Verified numerically (`code/ag5b_constant_check.py`): at even d the section $X + 1$ collapses in 22/22 samples with the top level never needed, while $c \in \{2, 3, -2, 5\}$ collapse only at the single degenerate point $(r, p) = (13, 5)$, where $\Phi_4(5) = 26$ has no prime outside $2r$ and the top level is vacuous for every section.

Proposition 5.10 (the $X + 1$ variant is mechanism-equivalent to Agrawal's conjecture)**.** *Suppose* $n \equiv p^j \pmod{\mathrm{lcm}(r, \rho_r(p))}$ *and* $p - 1 \mid n - 1$ *(the matching congruence plus the Carmichael condition, as in the original L–P sufficient construction). Then* $T(1, n, r)$ *holds modulo* p *outright. Consequently every counterexample to Agrawal's conjecture produced by the mechanism also defeats the variant conjecture whose hypothesis is the pair* $\{T(-1, n, r), T(1, n, r)\}$*: that variant, though formally stronger-looking, is exactly as resistant as Agrawal's — the second hypothesis adds no constraint within these assumptions. The section* $X+2$ *generally imposes a separate order condition; that is not a proof that it blocks every matching construction.*

Proof. At the $X = 1$ component, $(1 + 1)^n = 2^n \equiv 2 \pmod{p}$ since $\mathrm{ord}_p(2) \mid p - 1 \mid n - 1$. In the Φ_r components, $\zeta + 1 = (\zeta^2 - 1)/(\zeta - 1)$, a unit (its norm is $\Phi_r(-1) = 1$), and both $\zeta - 1$ and $\zeta^2 - 1$ are values of $X - 1$ at primitive r-th roots of unity, so their orders divide $\rho_r(p) = \mathrm{lcm}_a \, \mathrm{ord}(\zeta^a - 1)$. Hence $(\zeta^a - 1)^n = (\zeta^a - 1)^{p^j} = \zeta^{ap^j} - 1$ for $a = 1, 2$, and

$$(\zeta + 1)^n \;=\; \frac{(\zeta^2 - 1)^n}{(\zeta - 1)^n} \;=\; \frac{\zeta^{2p^j} - 1}{\zeta^{p^j} - 1} \;=\; \zeta^{p^j} + 1 \;=\; \zeta^n + 1,$$

using $n \equiv p^j \pmod{r}$ in the last step. Measured (`code/ag5b_constant_check.py`, check W): 48/48 synthetic mechanism instances pass $T(1, n, r)$, while 0/48 pass $T(2, n, r)$. □

5.1 Numerical validation

`code/ag5_popovych_check.py` (deterministic; output in `data/generated/ag5_popovych_check.json`) verifies, over 56 sample primes spanning the same (r, d) grid as Section 4:

check	statement	result
P1	Frobenius matching for $X+2$	56/56
P2	the $X-1$ collapse bound does *not* absorb $X+2$ (even d)	35/36
P3	top level $e=d$ present in $\rho'_r(p)$	55/56
P4	large primes of *all* levels $e \mid d$ divide $\rho'_r(p)$	47/47
P5	burden table: $\text{burden}_{\text{Pop}}$ vs. $\text{burden}_{\text{Agr}}$, 11 moduli	as stated
P6	$\gcd(r, 2^r+1) = 1$ (Lemma 5.2)	11/11
P7	heuristic Dickman cost table (Remark 5.7)	as stated

The single P2/P3 exception is $r=13$, $p=5$: there $\Phi_4(5) = 26 = 2 \cdot 13$ contains no prime outside $2r$ at all, so the top level is vacuous for this one tiny prime — a small-prime degeneracy, not a collapse event; it is consistent with (indeed predicted by) the generic model, under which exceptions have probability modeled historically as $\sim 1/\ell$ per large prime ℓ, with ≈ 0.28 misses estimated over this sample. This is not a verified probability law for the actual conjugate values. The burden deltas of P5: $+2$ at $r \in \{5, 17, 29, 41\}$, $+1$ at $r \in \{13, 37\}$, 0 at the $r \equiv 3 \pmod 4$ moduli (where the Agrawal burden was already ≥ 3 and both conjectures share full depth).

6 The impersonation hierarchy

Theorems 3.7 and 5.6 are statements about which congruence cells a counterexample mechanism forces its pool primes into. This section defines those cells once and for all, maps the classical generalized-Carmichael literature onto them, and records what is known about each — the related-work backbone and vocabulary for the negative-result paper (`doc/negative-result-plan.md`).

Definition 6.1 (impersonation cells)**.** *Let p be prime and n a positive integer. For a finite set S of levels and powers $\mathbf{i} = (i_e)_{e \in S}$, say n* impersonates Frobenius at p on S with powers $\mathbf{i}$ *if*

$$\Phi_e(p) \mid n - p^{i_e} \qquad \text{for every } e \in S.$$

By the dictionary of Section 2 this says exactly that $x^n = x^{p^{i_e}}$ for all x in the order-$\Phi_e(p)$ subgroup of $\mathbb{F}_{p^e}^\times$ — the points of the torus T_e. Uniform-power impersonation at depth m *is the single congruence $p^m - 1 \mid n - p^i$, i.e. $x^n = x^{p^i}$ for all $x \in \mathbb{F}_{p^m}^\times$. A squarefree composite n* lies in a cell *when every $p \mid n$ impersonates accordingly. Note two normalizations: at level 1 all powers coincide ($p \equiv 1 \bmod p-1$), so only the cell membership matters there; and on T_2 one has $p^i \equiv (-1)^i \pmod{p+1}$, so at level 2 only the* parity *of i_2 matters — the parity is a sign: even $\leftrightarrow$ target $+1$, odd $\leftrightarrow$ target -1.*

Proposition 6.2 (the classical objects in cell coordinates)**.** *For squarefree composite n:*

(i) n is a Carmichael number (Korselt: $p-1 \mid n-1$ for all $p \mid n$) iff n lies in the cell $(\{1\}, 0)$.

(ii) n is a Lucas–Carmichael number ($p+1 \mid n+1$) iff n lies in $(\{2\}, 1)$: indeed $p+1 \mid n-p \iff p+1 \mid n+1$.

(iii) n is a Williams number [14, 15, 16] (both conditions) iff n lies in $(\{1, 2\}, (0, 1))$, equivalently iff $n \equiv p \pmod{(p^2-1)/2}$ for every $p \mid n$.

(iv) *n is a Carmichael number of order m in the sense of Howe [17] ($x \mapsto x^n$ an endomorphism of every $\mathbb{Z}/n$-algebra generated as a $\mathbb{Z}/n$-module by at most m elements) iff for every $p \mid n$ and every depth $r \le m$ some power works: $\exists i,\ p^r - 1 \mid n - p^i$ (Howe's Theorem 1). For $m = 2$ this includes the exact congruence modulo $p^2 - 1$. Separate level-1 and level-2 congruences control only $\operatorname{lcm}(p-1, p+1) = (p^2-1)/2$ and can miss a factor of 2. For example, at $p = 7$, $n = 217$ satisfies $n \equiv 1$ modulo both 6 and 8, but $n \equiv 25 \pmod{48}$ is neither 1 nor 7, the powers of 7 modulo 48. Howe's* rigid *numbers are the all-powers-0 (all-even-parity) case $\operatorname{lcm}_{r \le m}(p^r - 1) \mid n - 1$.*

(v) *The Lenstra–Pomerance pool for Agrawal's conjecture is the Williams cell refined by the local conditions ($p \equiv 3 \bmod 80$, validity $n \not\equiv \pm 1 \bmod r$); Theorem 3.7 says the AKS mechanism at any modulus r forces the level set $\{e : e \mid d'\}$, with the Williams cell $\{1, 2\}$ as the unique dimension-2 minimum; Theorem 5.6 says the Popovych pair forces the full level set $\{e : e \mid d\}$. Grantham's Frobenius pseudoprimes [22] are the same impersonation idea tested against a single polynomial section rather than all torus points.*

Proof. (i)–(iii) unwind Definition 6.1 ((iii): $n \equiv 1 \bmod p - 1$ and $n \equiv -1 \equiv p \bmod p + 1$ combine by CRT to $n \equiv p \bmod \operatorname{lcm}(p-1, p+1) = (p^2-1)/2$); the point-identity form of the dictionary is verified numerically in Section 6.1. (iv) is Howe's Theorem 1 [17], retaining the full moduli. (v) collects the conditional support models in Sections 3 and 5. □

Theorem 6.3 (status of the cells — survey)**.** *(a)* *$(\{1\}, 0)$, Carmichael: infinitely many [13];* 105 *below 10^7.*

(b) *$(\{2\}, 1)$, Lucas–Carmichael: infinitely many — Wright [19], as a by-product of the infinitude of elliptic Carmichael numbers;* 135 *below 10^7.*

(c) *$(\{1, 2\}, (0, 1))$, Williams: open whether the cell is even nonempty. Williams [14] posed the question and showed such n need at least four prime factors; McIntosh–Mitra [16] verified none exist below $2^{64} \approx 1.8 \times 10^{19}$ and proved finiteness results (only finitely many with a given set of all but three prime factors). The Agrawal cell (v) is a refinement, a fortiori empty there. OEIS A329223 is a nonempty sequence of certain base-2 pseudoprimes; it is not the Williams-number sequence and does not itself supply this bound.*

(d) *Depth-2, any parity (Howe order 2):* nonempty*. Howe's explicit example $n = 443372888629441 = 17 \cdot 31 \cdot 41 \cdot 43 \cdot 89 \cdot 97 \cdot 167 \cdot 331$ is rigid of order 2 ($p^2 - 1 \mid n - 1$ for all eight primes; re-verified here, Section 6.1). Every order-m Carmichael number has at least $m + 2$ prime factors [18]; no order-3 example is known; Howe's heuristic predicts infinitely many of every order.*

(e) *Elliptic rungs: Silverman's elliptic Korselt criteria [20], conditional infinitude of elliptic Carmichael numbers by Ekstrom–Pomerance–Thakur [21], unconditional by Wright [19] — the structures item AG4 will build on.*

Remark 6.4 (sign splitting as a structural distinction). Contrast (c) and (d): rigid depth-2 impersonators exist at 4.4×10^{14}, while the Williams cell — the *same* level set — is empty to 2^{64}. This comparison suggests a sign-structure distinction; it does not prove a constructibility boundary. The rigid cell's condition is one-sided: $n \equiv 1 \pmod{\operatorname{lcm}_{p \mid n}(p^2 - 1)}$ is an AGP-type problem — harvest primes with $p^2 - 1 \mid \Lambda$ for a single highly composite Λ (the $p - 1$ and $p + 1$ parts may share prime support freely) and solve one subset-product with target 1; that machine produced (d) at modest size. The Williams parities $(0, 1)$ instead force target $+1$ on the level-1 side and -1 on the level-2 side; on a shared odd prime power the two targets conflict, so the sides must be coprime

away from 2 — exactly the side-coprimality (working paper Lemma 2.3) whose harvest collapse this project measured as N1 and whose two-sided solver obstruction is A10. In cell language: the project's original target is a *sign-split* cell, whereas Howe's example shows that an unsplit cell is reachable. N1/A10 are finite experiments with modeling limitations, not a proof that the sign-split cell is unconstructible.

For the original $r = 5$ *full-bound* construction, rigidity $p^2 - 1 \mid n - 1$ is incompatible with an odd Frobenius exponent: $n \equiv 1 \pmod{p+1}$ but $p^j \equiv -1 \pmod{p+1}$ for odd j. An even exponent gives $n \equiv \pm 1 \pmod 5$, inside the exception clause. This argument uses the full-bound design and is not an exclusion of all rigid cells for every modulus or actual section order. Concretely: Howe's rigid n has $n \equiv 1 \pmod 5$, yet it does not even pass $T(-1, n, 5)$ (Section 6.1): rigidity controls the $p^2 - 1$ part of $\rho_5(p)$ but not the extra μ_{2r}-torsion — the factor $2r$ in Lemma 3.2's bound — which is exactly the 2-adic bookkeeping the L–P conditions modulo 80 exist to fix.

Remark 6.5 (imports for the negative-result paper). Three direct imports: (1) *vocabulary* — the paper's target objects are Williams numbers in prescribed congruence cells, an established name with a 1977 pedigree; (2) *search floor* — McIntosh–Mitra's 2^{64} bound is the modern version of the Pinch-table emptiness the README cites; (3) *calibration* — Howe's construction of (d) is the nearest success of the AGP machine to our target cell, and the one-sided/sign-split contrast of Remark 6.4 is the structural explanation of why his succeeded where H8/N1 collapsed; any future attempt on the Williams cell must engage that contrast, not the depth.

6.1 Numerical checks

`code/ag3_hierarchy_check.py` (deterministic; output in `data/generated/ag3_hierarchy_check.json`):

check	statement	result
H1	Howe's n is squarefree, order 2 and rigid; not Williams	confirmed
H2	dictionary on T_2: congruence $\iff$ point identity, random (p, n, i)	60/60
H3	census below 10^7: Carmichael / Lucas–C / Williams / order-2	105 / 135 / 0 / 0
H4	Howe's rigid n passes $T(-1, n, 5)$	**no**

In H1 the matching level-2 powers at every p are $\{0, 2, 4\}$ — all even, the rigid parity; no prime is in the Lucas cell, placing n outside Williams as expected. H3's Carmichael count (105 below 10^7) matches the known value, validating the census code; the smallest members found are 561 (Carmichael) and 399 (Lucas–Carmichael). H4 realizes Remark 6.4's last point: even within the exception clause, rigid impersonation alone does not pass the AKS congruence — the μ_{2r} torsion must be arranged separately.

7 The elliptic axis: formulation and the stabilizer-index floor

Item AG4 asks whether elliptic torsion provides an alternative construction. The cofactor identities below are useful upper bounds. They do not prove generic support saturation, a mandatory exception clause, or transfer of an algorithmic obstruction from the torus experiments.

7.1 Formulation (AG4a): the test needs CM

The AKS congruence rests on the *canonical* substitution $\sigma_n : X \mapsto X^n$ — the Adams operation of the Λ-ring $\mathbb{Z}[\mu_r]$ (Section 1). If an elliptic curve carried such a canonical commuting family, each

ψ_p would be an endomorphism of degree p lifting Frobenius, and an endomorphism of non-square degree already forces $\mathrm{End}(E) \supsetneq \mathbb{Z}$, i.e. complex multiplication. This restricts that particular endomorphism-lift formulation, not all elliptic primality tests. Gurney's thesis [23] studies CM moduli and curves of Shimura type with Λ-structures, with base and descent hypotheses; it is not a blanket assertion for every CM curve over $\mathbb{Q}$. We therefore consider only the following ordinary, split CM setting:

Definition 7.1 (elliptic AKS test, CM form)**.** *Let $E/\mathbb{Q}$ have CM by O_K, K imaginary quadratic with unit group $O_K^\times$ of order w ($w = 4$: $j = 1728$, $E : y^2 = x^3 - x$, $[i](x, y) = (-x, iy)$; $w = 6$: $j = 0$). Fix an odd prime r of good reduction split in K, a kernel polynomial g of an r-eigenline (certificate-style, or explicit for special sequences as in [10]), and a section f (we take $f = x$). The test: for composite-candidate n coprime to $2r\,\Delta_E$,*

$$\exists\, \tau \in (\mathbb{Z}/r)^\times : \qquad f^{\,n} \;\equiv\; f \circ [\tau] \pmod{n,\ g(x)},$$

$[\tau]$ acting through division polynomials ($r - 1$ candidates, so poly(r) trials). For n prime, ordinary and split, the n-power Frobenius acts on the eigenline by a scalar λ_n and the test passes with $\tau = \lambda_n$.

Remark 7.2 (the exception clause). Let $U \le (\mathbb{Z}/r)^\times$ be the image of $O_K^\times$ (for $j = 1728$: $U = \mu_4 = \{\pm 1, \pm i\}$, present iff $4 \mid r - 1$). Cells with $\lambda_p \in U$ at every $p \mid n$ are the elliptic analogue of the rigid cells of Section 6: their assembly is a one-sided AGP-type problem in analogy with the rigid torus cell. This analogy does not prove that an elliptic conjecture must adopt $\tau \in U$ as its exception. For the following index model we *choose* that exception and restrict to Frobenius-matching cells escaping U. The need for that exception, and any construction of composite examples inside it, remain separate questions.

7.2 The stabilizer-index theorem (AG4b)

The collapse identities are exact, as in Lemma 3.2: for an eigenline point T with Frobenius scalar λ, $x(T)^{p^e} = x([\lambda^e]T)$, and on $E : y^2 = x^3 - x$ the automorphism gives $x([i]S) = -x(S)$ for *all* S. Hence $\lambda^e \equiv \pm 1$ gives $x(T)^{p^e} = x(T)$ and $\lambda^e \equiv \pm i$ gives $x(T)^{p^e} = -x(T)$; with

$$k' \;:=\; \min\{\, e \ge 1 : \lambda^e \in U \,\} \;=\; [\langle\lambda\rangle : \langle\lambda\rangle \cap U],$$

the order of $x(T)$ divides $2(p^{k'} - 1)$. This is a containing support bound, not an attained order or a necessary smoothness burden.

Theorem 7.3 (cofactor-based stabilizer-index upper bound)**.** *Let a test section f on a torsion scheme carry a scalar Frobenius action λ in a cyclic group Γ. Suppose $f(T) \neq 0$, and $U \le \Gamma$ has exact cofactor identities after good reduction: $f([u]T) = \eta_u(T) f(T)$, where $\eta_u(T)^L = 1$ for a specified L. Set $k' = [\langle\lambda\rangle : \langle\lambda\rangle \cap U]$. Then*

$$\mathrm{ord}(f(T)) \mid L(p^{k'} - 1).$$

If the chosen model additionally requires $\lambda \notin U$, its index k' is at least 2. The index is $d/\gcd(d, 2)$ for $U = \{\pm 1\}$, d for $U = \{1\}$, and $m/\gcd(m, w)$ when $\mathrm{ord}(\lambda) = m$ and U is a subgroup of order w in the cyclic scalar group. Attainment of all large-prime support in this upper bound is a separate section-specific hypothesis, not a conclusion of the index formula.

Proof. The index is the order of λ in $\langle\lambda\rangle/(\langle\lambda\rangle \cap U)$, so $\lambda^{k'} \in U$. Frobenius and the assumed cofactor identity give $f(T)^{p^{k'}} = \eta_{\lambda^{k'}}(T) f(T)$. Dividing by the nonzero value and raising to L proves the order bound. The cyclic-group index formulas and the inequality for $\lambda \notin U$ follow directly. No assertion of minimal finite-field collapse or disjoint elliptic prime supports is used. □

Remark 7.4 (supersingular cells). For p inert in K (e.g. $p \equiv 3 \bmod 4$ on $y^2 = x^3 - x$), $a_p = 0$ and $\varphi^2 = [-p]$, so the collapse of x-values is governed by $\langle p \rangle \subseteq (\mathbb{Z}/r)^\times$ exactly as in the torus case at the level of this endomorphism relation. A complete supersingular matching and assembly analysis is not supplied by that relation.

7.3 Gate AG-G2: not passed

Remark 7.5 (conditional elliptic index model; axis not closed). On the $j = 1728$ curve, the arithmetic candidate for the containing index is $\mathrm{burden}_{\mathrm{ell}}(r) = \min\{\, m/\gcd(m,4) \geq 2 : m \mid r-1 \,\}$, with no index outside U at $r = 5$; for the analogous $j = 0$ model replace 4 by 6, where the specified unit image exists. In the historical table (Section 7.4) $\mathrm{burden}_{\mathrm{ell}}(r) \geq \mathrm{burden}_{\mathrm{Agr}}(r)$, and index 2 is arithmetically available iff $8 \mid r-1$ ($j = 1728$) or $12 \mid r-1$ ($j = 0$). Its containing support is $\{1, 2\}$. Interpreting this as a necessary twin-smooth supply condition requires elliptic support saturation and a specified exception/matching model. Neither the existence of appropriate harvested eigenline data nor transfer of an algorithmic obstruction has been proved. The historical decision not to pursue AG4(c) is not a mathematical closure of the elliptic axis.

Caveats, beyond the standing ones (mechanism scope, Remark 2.3; generic sharpness, Remark 3.10): the chosen endomorphism-lift formulation is CM-bound, without excluding other elliptic tests, and the eigenline kernel polynomial is certificate-style data for general n (explicit only for special sequences, as in [10, 24]).

7.4 Numerical validation

`code/ag4_elliptic_check.py` (deterministic; output in `data/generated/ag4_elliptic_check.json`), on $E : y^2 = x^3 - x$ with $r \in \{13, 17\}$ and ten ordinary primes $p \equiv 1 \pmod 4$ each:

check	statement	result
E1	$\deg \psi_r = (r^2-1)/2$; factor degrees multiply back	all
E2	Frobenius closure $t^{p^k} = t$ per factor field	all
E3	exceptional cells ($k' = 1$, $\lambda \in \mu_4$) have $k \leq 2$	45/45
E3	sharpness: large primes of levels $e \mid k'$ divide $\mathrm{ord}(t)$	112/112
E4	measured valid-cell floor: $r = 13 \to 3$, $r = 17 \to 2$	matches index formula
E5	$\mathrm{burden}_{\mathrm{ell}}(r) \geq \mathrm{burden}_{\mathrm{Agr}}(r)$ over the A11 moduli	11/11

The $r = 17$ data shows the floor concretely: degree-4 factors with $t^{p^2} = -t$ (order-8 λ: burden 2, $\Phi_4(p)$ absent from $\mathrm{ord}\, t$) alongside degree-8 factors with $k' = 4$ (order-16 λ: levels $\{1, 2, 4\}$) and degree-16 mixed cells; at $r = 13$ no valid cell beats burden 3, as $12/\gcd(12, 4) = 3$ predicts. $r = 5$ is degenerate on this curve: every λ-order divides 4, so all cells are exceptional.

A A function-field analogy, not a sharpness theorem

Over $A = \mathbb{F}_q[t]$ the Carlitz module $C : A \to \mathrm{End}(\mathbb{G}_a)$ has $C_t(x) = tx + x^q$ and commuting operations C_a. For monic irreducible P of degree d its standard Frobenius congruence is $C_P(x) \equiv x^{q^d} \pmod P$. For an auxiliary monic irreducible R, $A[x]/(C_R(x))$ describes the full torsion algebra, including the zero point; the nonzero torsion cyclotomic polynomial is $C_R(x)/x$. These give natural analogies for Frobenius data and cofactor bounds, not an automatic transport of all integer support and assembly statements.

Function-field Chebotarev and Weil's bounds provide equidistribution for specified finite extensions, with errors depending on the extension and its constant field and genus. To deduce the sharpness or smooth-order supply claims considered here, one must specify those extensions, the asymptotic regime, the varying order primes, uniform error bounds, local obstructions, and the joint smoothness conditions. This appendix supplies none of those arguments. Consequently it does *not* prove Hypothesis 3.5, infinitude of a proposed smooth pool, or an unconditional function-field burden lower bound.

Irreducibility testing over a finite field is deterministic polynomial time via modular Frobenius powers and gcd tests. That fact does not establish the proposed counting assertions or identify the integer problem's sole remaining difficulty as an equidistribution gap. The analogy remains an exploratory direction.

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